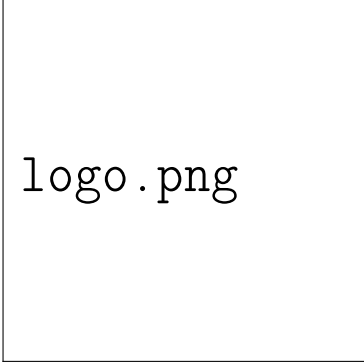




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Comprehensive Wind Energy Systems

and System Advisor Model (SAM) Training

Formation Course for Energy Engineering Students

POWERUP Club
École Nationale des Sciences Appliquées de Fès
Energy Engineering and Smart Systems Department

November 17, 2025

Preface

Welcome to the comprehensive Wind Energy Systems and System Advisor Model (SAM) training course, developed by the POWERUP Club at ENSA Fes. This course is designed specifically for students of Energy Engineering and Intelligent Systems (Génie Énergétique et Systèmes Intelligents) as part of our ongoing series of simulation software training programs.

About POWERUP Club

POWERUP is a student club dedicated to advancing knowledge and practical skills in renewable energy systems, with a special focus on wind and solar power. This formation represents the second major training session of the year, following our series on renewable energy simulation tools.

Course Objectives

This comprehensive course aims to provide students with:

- A solid theoretical foundation in wind energy systems
- Practical understanding of wind resource assessment
- Knowledge of wind turbine aerodynamics and components
- Skills in wind farm design and optimization
- Hands-on experience with System Advisor Model (SAM) software
- Capabilities to perform techno-economic analysis of wind projects

How to Use This Course

This document is structured in 11 comprehensive chapters, progressing from fundamental concepts to advanced modeling techniques. Each chapter builds upon previous knowledge, culminating in practical SAM software training. We recommend following the chapters sequentially for maximum benefit.

The POWERUP Team
ENSA Fes, Morocco

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Chapter 1

Introduction to Wind Energy

1.1 Overview of Renewable Energy and Wind Power

Wind energy represents one of the fastest-growing renewable energy sources globally. As the world transitions toward sustainable energy systems, wind power has emerged as a critical component of the global energy mix. Wind energy harnesses the kinetic energy of moving air masses to generate electricity through wind turbines.

The fundamental principle behind wind energy is the conversion of atmospheric air movement into mechanical rotation, which subsequently drives electrical generators. This conversion process is clean, renewable, and produces zero direct greenhouse gas emissions during operation.

1.1.1 Historical Context

Wind power has been utilized by humanity for millennia, from ancient sailing vessels to traditional windmills for grain grinding and water pumping. The modern era of wind energy began in the late 19th century with the first electricity-generating wind turbines. However, large-scale commercial wind power development accelerated dramatically in the 1970s following the oil crisis.

Today's wind turbines bear little resemblance to their historical predecessors. Modern wind turbines are sophisticated machines incorporating advanced materials, aerodynamic designs, and intelligent control systems, capable of generating several megawatts of electrical power.

1.2 Global Wind Energy Production Trends

Global wind energy capacity has experienced exponential growth over the past two decades. According to recent data, worldwide wind power production has increased from approximately 221 TWh in 2008 to over 3,450 TWh in 2024, representing a compound annual growth rate exceeding 15%.

Table 1.1: Global Wind Energy Production (2008-2024)

Year	Production (TWh)
2008	221
2010	340
2012	489
2015	830
2020	1,590
2023	2,304
2024	3,450

This remarkable growth trajectory reflects technological improvements, cost reductions, favorable policy frameworks, and increasing global commitment to decarbonization. Major markets include China, the United States, Germany, India, and Spain, with emerging markets rapidly expanding their wind energy portfolios.

1.3 Wind Energy in Morocco - National Strategy

Morocco has positioned itself as a regional leader in renewable energy development, with wind power playing a central role in the country's energy transition strategy. The Kingdom has made substantial investments in wind energy infrastructure as part of its ambitious renewable energy targets.

1.3.1 Morocco's Energy Transition Vision

Morocco's energy strategy aims to achieve:

- **35% renewable electricity** by 2025
- **40% renewable electricity** by 2030
- **5,000 MW** of installed wind capacity by 2030

As of recent data, Morocco has already installed approximately 2,370 MW of wind power capacity, demonstrating significant progress toward these goals. This capacity is distributed across multiple wind farms throughout the country, particularly in regions with excellent wind resources.

1.4 Major Wind Farms in Morocco (Case Studies)

Morocco's wind energy landscape includes numerous large-scale wind farms developed by both public and private sector entities. The following table presents the major operational wind farms across the country:

Table 1.2: Major Wind Farms in Morocco

Wind Farm	Year	Capacity (MW)	Location	Developer
Tarfaya	2014	301	Tarfaya	NAREVA/ENGIE
Midelt	2020	210	Midelt	Masen
Khalladi	2017	120	Melloussa	Acwa Power/TAQA
Akhfennir 1	2013	101.87	Akhfennir	NAREVA
Akhfennir 2	2016	95.2	Akhfennir	NAREVA
Tanger I	2011	140	Tanger	ONEE
Boujdour	2023	300	Boujdour	NAREVA
Aftissat I	2019	201.6	Boujdour	NAREVA
Aftissat II	2023	200	Boujdour	NAREVA
Taza	2022	87	Taza	EDF/Mitsui
Essaouira	2006	60	Essaouira	ONEE

1.4.1 Case Study: Tarfaya Wind Farm

The Tarfaya Wind Farm represents Morocco's largest wind energy installation with 301 MW capacity. Located in southern Morocco, this facility benefits from exceptional wind resources along the Atlantic coast. The project showcases Morocco's capability to develop large-scale renewable energy infrastructure and demonstrates the country's commitment to clean energy.

1.5 Morocco's Energy Mix Goals (2025-2030)

Morocco's energy transition roadmap establishes clear targets for renewable energy integration into the national electricity mix:

- **2025 Target:** 35% of electricity generation from renewable sources
- **2030 Target:** 40% of electricity generation from renewable sources (with potential for 52%)
- **Wind Component:** 5,000 MW installed capacity by 2030

These objectives position Morocco among the most ambitious renewable energy programs globally. The strategy encompasses wind, solar, and hydroelectric resources, with wind energy contributing significantly to achieving energy independence and reducing carbon emissions.

Chapter 2

Wind Origin and Atmospheric Phenomena

2.1 Solar Radiation and Uneven Heating

Wind is fundamentally a result of differential solar heating of Earth's surface. The sun's radiation strikes different parts of Earth's surface at varying angles and intensities, creating temperature gradients that drive atmospheric circulation.

The primary mechanisms include:

2.1.1 Equatorial vs. Polar Heating

The equatorial regions receive more direct solar radiation throughout the year compared to polar regions. This creates a significant temperature differential between the equator and poles, establishing the primary driver for global wind patterns.

2.1.2 Land-Water Heating Differences

Land surfaces heat and cool more rapidly than water bodies due to differences in specific heat capacity. This differential heating creates localized wind patterns, particularly in coastal regions.

The mathematical relationship between solar irradiance and surface heating can be expressed as:

$$Q = I \cdot A \cdot \cos(\theta) \cdot (1 - \alpha)$$

where:

- Q = Heat absorbed by surface
- I = Solar irradiance
- A = Surface area
- θ = Angle of incidence
- α = Surface albedo (reflectivity)

2.2 Pressure Gradients and Wind Formation

Wind results from air movement from high-pressure regions to low-pressure regions. Pressure differences arise from temperature variations, as described by the ideal gas law:

$$P = \rho RT$$

where:

- P = Pressure
- ρ = Air density
- R = Specific gas constant for air
- T = Temperature

The pressure gradient force (PGF) can be expressed as:

$$\text{PGF} = -\frac{1}{\rho} \nabla P$$

This force accelerates air from high to low pressure, creating wind. The magnitude of wind velocity correlates with the pressure gradient strength.

2.3 Daily Wind Phenomena

2.3.1 Sea and Land Breezes

Sea and land breezes represent classic examples of thermally-driven local wind systems resulting from differential heating between land and water surfaces.

Sea Breeze (Daytime)

During daytime:

1. Land heats faster than the ocean
2. Warm air over land rises (creating low pressure)
3. Cooler air from the ocean flows inland
4. Onshore wind develops (sea breeze)

Land Breeze (Nighttime)

During nighttime:

1. Land cools faster than the ocean
2. Air over ocean remains warmer and rises
3. Cooler air from land flows toward ocean
4. Offshore wind develops (land breeze)

These phenomena are particularly significant for coastal wind energy projects, as they create predictable diurnal wind patterns.

2.3.2 Mountain and Valley Winds

Mountain-valley wind systems result from differential heating of slopes and valley floors:

Valley Winds (Anabatic)

During daytime, sun-facing slopes heat rapidly, warming adjacent air which rises up the slope, creating upslope winds (anabatic winds).

Mountain Winds (Katabatic)

During nighttime, slopes cool through radiative heat loss, cooling adjacent air which becomes denser and flows downslope (katabatic winds).

2.4 Coriolis Effect and Global Wind Patterns

The Coriolis effect results from Earth's rotation and significantly influences large-scale wind patterns. Moving air masses experience an apparent deflection:

- **Northern Hemisphere:** Deflection to the right
- **Southern Hemisphere:** Deflection to the left

The Coriolis parameter is given by:

$$f = 2\Omega \sin(\phi)$$

where:

- f = Coriolis parameter
- Ω = Earth's angular velocity (7.29×10^{-5} rad/s)
- ϕ = Latitude

This effect creates the characteristic global wind patterns including trade winds, westerlies, and polar easterlies.

2.5 Wind Classification by Scale

Atmospheric motions span multiple spatial and temporal scales:

Table 2.1: Wind Classification by Scale

Scale	Spatial Extent	Examples
Planetary	1,000+ km	Trade winds, jet streams
Synoptic	100-1,000 km	High/low pressure systems
Mesoscale	1-100 km	Sea breezes, thunderstorms
Microscale	<1 km	Turbulence, gusts

For wind energy applications, understanding mesoscale and microscale phenomena is particularly important for resource assessment and turbine siting.

Chapter 3

Wind Characteristics and Resource Assessment

3.1 Turbulence and Flow Regimes

Wind flow exhibits varying degrees of turbulence depending on terrain characteristics, atmospheric stability, and height above ground. Understanding turbulence is critical for wind turbine design and performance prediction.

3.1.1 Laminar vs. Turbulent Flow

Flow regimes are characterized by the Reynolds number:

$$Re = \frac{\rho V L}{\mu}$$

where:

- Re = Reynolds number
- ρ = Air density
- V = Wind velocity
- L = Characteristic length
- μ = Dynamic viscosity

Laminar flow ($Re < 2300$): Smooth, layered flow with minimal mixing

Turbulent flow ($Re > 4000$): Chaotic flow with significant mixing and eddies

Atmospheric flows are almost always turbulent, with turbulence intensity varying significantly based on surface roughness and atmospheric conditions.

3.1.2 Turbulence Intensity

Turbulence intensity (TI) quantifies wind speed fluctuations:

$$TI = \frac{\sigma_v}{\bar{v}}$$

where:

- σ_v = Standard deviation of wind speed
- $\bar{v}$ = Mean wind speed

Typical values:

- Offshore/smooth terrain: TI = 0.10-0.15
- Onshore/moderate terrain: TI = 0.15-0.25
- Complex terrain: TI > 0.25

High turbulence increases mechanical loading on turbines and can reduce energy production.

3.2 Wind Shear and Vertical Wind Profile

Wind speed increases with height above ground due to reduced surface friction. This vertical wind profile is fundamental to wind energy, as turbine hub heights are selected to access higher wind speeds.

3.2.1 Power Law Model

The power law describes wind speed variation with height:

$$\frac{v(z)}{v(z_{ref})} = \left(\frac{z}{z_{ref}} \right)^\alpha$$

where:

- $v(z)$ = Wind speed at height z
- $v(z_{ref})$ = Wind speed at reference height z_{ref}
- α = Power law exponent (shear coefficient)

3.2.2 Logarithmic Profile

The logarithmic law provides a more physically-based description:

$$v(z) = \frac{u_*}{\kappa} \ln \left(\frac{z}{z_0} \right)$$

where:

- u_* = Friction velocity
- κ = von Kármán constant (0.4)
- z_0 = Surface roughness length

3.3 Power Law and Logarithmic Profile

3.3.1 Shear Coefficient Values

The power law exponent α depends on surface roughness and atmospheric stability:

Table 3.1: Typical Shear Coefficient Values

Terrain Type	α
Smooth water surface	0.10
Open grassland	0.14
Agricultural land	0.20
Urban/forest	0.30-0.40

3.4 Terrain Effects and Surface Roughness

Surface roughness length (z_0) quantifies terrain resistance to air flow:

Table 3.2: Surface Roughness Length by Terrain

Terrain Type	z_0 (m)
Water surface	0.0002
Short grass	0.01
Crops	0.05
Scattered trees	0.10
Forest	0.50-1.00
Urban areas	1.00-2.00

Terrain features significantly affect wind resource quality. Complex topography creates:

- Speed-up effects on ridges
- Sheltering in valleys
- Flow separation and turbulence
- Channeling through passes

3.5 Wind Speed Distribution

Wind speed at a given location varies continuously. Statistical distributions characterize this variability, enabling long-term energy production estimates.

3.6 Weibull Distribution Analysis

The Weibull distribution is the standard model for wind speed probability:

$$f(v) = \frac{k}{c} \left(\frac{v}{c}\right)^{k-1} \exp \left[- \left(\frac{v}{c}\right)^k \right]$$

where:

- $f(v)$ = Probability density function
- k = Shape parameter (dimensionless)
- c = Scale parameter (m/s)
- v = Wind speed (m/s)

3.6.1 Weibull Parameters

Shape parameter (k):

- $k < 2$: Highly variable winds
- $k = 2$: Rayleigh distribution (common approximation)
- $k > 2$: More consistent winds

Scale parameter (c):

- Related to mean wind speed
- Approximately: $c \approx 1.13 \times \bar{v}$ for $k = 2$

3.6.2 Mean Wind Speed from Weibull

$$\bar{v} = c \cdot \Gamma \left(1 + \frac{1}{k} \right)$$

where Γ is the gamma function.

3.6.3 Parameter Estimation

Common methods for determining Weibull parameters:

1. **Empirical Method of Lysen**
2. **Maximum Likelihood Method**
3. **Method of Moments**
4. **Graphical Method**

The Weibull distribution enables calculation of wind power density and annual energy production without requiring hour-by-hour simulation.

Chapter 4

Wind Turbine Aerodynamics

4.1 Fundamental Physics of Wind Energy

Wind energy conversion involves extracting kinetic energy from moving air masses. Understanding the fundamental physics governing this process is essential for turbine design and performance analysis.

4.2 Kinetic Energy and Wind Power

The kinetic energy in wind flowing through an area is:

$$E_{kinetic} = \frac{1}{2}mv^2$$

where:

- m = Mass of air
- v = Wind velocity

For continuous flow through a swept area A , the power (energy per unit time) is:

$$P_{wind} = \frac{1}{2}\rho Av^3$$

where:

- ρ = Air density (kg/m³)
- A = Swept area (m²)
- v = Wind speed (m/s)

4.2.1 Key Insight: Cubic Relationship

The cubic relationship between power and wind speed is fundamental:

- Doubling wind speed increases power by factor of 8
- Small increases in wind speed yield large power gains
- This relationship drives the value of accurate resource assessment

4.2.2 Air Density Variations

Air density depends on temperature, pressure, and humidity:

$$\rho = \frac{P}{R \cdot T}$$

Standard conditions: $\rho = 1.225 \text{ kg/m}^3$ at sea level, 15°C

Variations due to:

- Altitude (density decreases 12% per 1000m)
- Temperature (density decreases with heating)
- Humidity (moist air less dense than dry air)

4.3 Actuator Disk Theory

Actuator disk theory models the rotor as a permeable disk that extracts energy from the wind. This simplified representation enables fundamental analysis of wind turbine performance.

4.3.1 Control Volume Analysis

Consider a streamtube passing through the rotor disk:

- Upstream: Undisturbed wind speed v_0
- At rotor: Wind speed v_r
- Downstream: Wake wind speed v_w

The rotor slows the wind, expanding the streamtube to maintain mass continuity.

4.4 Conservation of Mass and Momentum

4.4.1 Mass Conservation

Mass flow rate through the system remains constant:

$$\dot{m} = \rho A_0 v_0 = \rho A_r v_r = \rho A_w v_w$$

4.4.2 Momentum Conservation

The thrust force on the rotor equals the rate of change of momentum:

$$T = \dot{m}(v_0 - v_w) = \rho A_r v_r (v_0 - v_w)$$

4.4.3 Axial Induction Factor

The axial induction factor a quantifies flow deceleration:

$$a = \frac{v_0 - v_r}{v_0}$$

Therefore:

$$v_r = v_0(1 - a)$$

Wake velocity:

$$v_w = v_0(1 - 2a)$$

4.5 Bernoulli's Equation Application

Applying Bernoulli's equation upstream and downstream of the rotor:

$$P_0 + \frac{1}{2}\rho v_0^2 = P_1 + \frac{1}{2}\rho v_r^2$$

$$P_2 + \frac{1}{2}\rho v_r^2 = P_w + \frac{1}{2}\rho v_w^2$$

The pressure difference across the disk:

$$\Delta P = P_1 - P_2 = \frac{1}{2}\rho(v_0^2 - v_w^2)$$

4.6 Betz's Limit - Maximum Power Extraction

The power extracted by the rotor:

$$P = T \cdot v_r = \rho A_r v_r (v_0 - v_w) \cdot v_r$$

Substituting relationships in terms of a :

$$P = \frac{1}{2}\rho A v_0^3 \cdot 4a(1 - a)^2$$

4.6.1 Power Coefficient

The power coefficient C_P is the ratio of extracted power to available wind power:

$$C_P = \frac{P}{P_{wind}} = 4a(1 - a)^2$$

4.6.2 Maximum Theoretical Efficiency

To find maximum C_P , differentiate with respect to a :

$$\frac{dC_P}{da} = 4(1 - a)^2 - 8a(1 - a) = 0$$

Solving: $a = 1/3$

Therefore:

$$C_{P,max} = \frac{16}{27} \approx 0.593$$

Betz's Limit: No wind turbine can convert more than 59.3% of wind's kinetic energy into mechanical power.

4.7 Power Coefficient (C_P) and Efficiency

Real wind turbines achieve C_P values of 0.35-0.50 due to:

- Aerodynamic losses (drag, tip losses)
- Mechanical losses (gearbox, bearings)
- Electrical losses (generator, power electronics)
- Non-optimal operation at varying wind speeds

The power coefficient depends on:

$$C_P = f(\lambda, \beta)$$

where:

- λ = Tip speed ratio
- β = Blade pitch angle

4.7.1 Tip Speed Ratio

$$\lambda = \frac{\omega R}{v}$$

where:

- ω = Rotor angular velocity (rad/s)
- R = Rotor radius (m)
- v = Wind speed (m/s)

Modern horizontal-axis wind turbines operate at $\lambda = 6 - 8$ for optimal performance.

Chapter 5

Wind Turbine Systems and Components

5.1 Wind Turbine Types (HAWT vs VAWT)

Wind turbines are classified by rotor axis orientation:

5.1.1 Horizontal-Axis Wind Turbines (HAWT)

Characteristics:

- Rotor axis parallel to ground
- Most common commercial configuration
- Typically 3-bladed upwind design
- Requires yaw mechanism
- Higher efficiency than VAWT

Advantages:

- High power coefficient (0.4-0.5)
- Mature technology
- Extensive operational experience
- Scalable to multi-MW sizes

Disadvantages:

- Complex yaw control
- Tower shadow effects
- Height-dependent installation challenges

5.1.2 Vertical-Axis Wind Turbines (VAWT)

Characteristics:

- Rotor axis perpendicular to ground
- Examples: Darrieus, Savonius designs
- Omnidirectional (no yaw required)
- Lower efficiency than HAWT

Advantages:

- No yaw mechanism needed
- Generator at ground level
- Simpler maintenance access
- Better for turbulent urban environments

Disadvantages:

- Lower power coefficient (0.25-0.35)
- Limited commercial deployment
- Structural fatigue issues
- Starting torque limitations

5.2 Rotor Design and Blade Aerodynamics

The rotor is the primary energy-capturing component, with design directly impacting performance.

5.2.1 Number of Blades

Table 5.1: Blade Configuration Comparison

Parameter	2-Blade	3-Blade	Multi-Blade
Solidity	Low	Medium	High
Optimal TSR	8-10	6-8	3-5
Efficiency	High	High	Medium
Cost	Lower	Higher	Highest
Visual impact	Lower	Better	Higher

Modern commercial turbines predominantly use 3-blade configuration, balancing efficiency, cost, and aesthetics.

5.2.2 Blade Aerodynamic Forces

Airfoils generate:

- **Lift force:** Perpendicular to relative wind, drives rotation
- **Drag force:** Parallel to relative wind, resists rotation

Lift coefficient:

$$C_L = \frac{L}{\frac{1}{2}\rho v^2 A}$$

Drag coefficient:

$$C_D = \frac{D}{\frac{1}{2}\rho v^2 A}$$

High C_L/C_D ratio essential for efficient energy extraction.

5.3 Airfoil Selection and Blade Geometry

5.3.1 Airfoil Characteristics

Wind turbine airfoils require:

- High lift-to-drag ratio
- Gentle stall characteristics
- Insensitivity to roughness (leading edge erosion)
- Structural strength
- Manufacturing feasibility

Common airfoil families:

- NACA 44xx series
- NREL S-series
- Delft University (DU) series
- Risø series

5.3.2 Blade Planform Design

Blade design involves specifying chord and twist distributions along the span to optimize performance.

Chord distribution: Typically wider at root, tapering toward tip

Twist distribution: Greater twist at root, decreasing toward tip, ensuring optimal angle of attack across blade span

5.4 Mechanical Components

5.4.1 Hub and Rotor Assembly

The hub connects blades to the main shaft, transmitting aerodynamic torque. Hub designs include:

- **Rigid hubs:** Fixed blade attachment
- **Teetering hubs:** Allows rotor tilting (2-blade turbines)
- **Pitch-controlled hubs:** Individual blade pitch adjustment

5.4.2 Gearbox and Transmission

Most turbines employ gearboxes to convert low-speed, high-torque rotor rotation to high-speed, low-torque generator input.

Typical specifications:

- Input: 10-20 rpm
- Output: 1200-1800 rpm
- Gear ratio: 60:1 to 120:1
- Efficiency: 95-98%

Gearbox types:

- Parallel shaft
- Planetary (most common)
- Hybrid configurations

Direct-drive alternatives: Some modern turbines eliminate gearboxes, using low-speed generators with higher pole counts, reducing mechanical complexity and maintenance.

5.4.3 Tower and Foundation

Tower types:

- Tubular steel (most common for large turbines)
- Lattice/truss structures
- Hybrid designs
- Concrete towers (very large turbines)

Foundation types:

- Gravity foundations (onshore)

- Pile foundations
- Rock-anchored foundations
- Floating platforms (offshore)

Tower height selection balances:

- Higher wind speeds at greater heights
- Increased structural costs
- Transportation and installation constraints

5.5 Electrical Components

5.5.1 Synchronous Generators

Characteristics:

- Rotor speed synchronized with grid frequency
- Requires external excitation
- Can provide reactive power control
- More expensive than induction generators

Types:

- Electrically excited synchronous generators (EESG)
- Permanent magnet synchronous generators (PMSG)

5.5.2 Asynchronous (Induction) Generators

Characteristics:

- Simple, robust construction
- Lower cost
- Slightly lower efficiency than synchronous
- Requires reactive power compensation

Configurations:

- Fixed-speed induction generator (FSIG)
- Doubly-fed induction generator (DFIG) - most common

5.5.3 Power Electronics and Converters

Modern variable-speed turbines employ power electronic converters:

Functions:

- Decouple rotor speed from grid frequency
- Enable maximum power point tracking
- Provide grid support capabilities
- Smooth power output

Converter topologies:

- Partial-scale converters (DFIG): 30% of rated power
- Full-scale converters (PMSG, EESG): 100% of rated power

Typical efficiency: >97%

5.6 Control Systems and Safety Features

5.6.1 Primary Control Objectives

1. **Power optimization:** Maximize energy capture below rated wind speed
2. **Power regulation:** Limit power at rated level above rated wind speed
3. **Load mitigation:** Reduce structural loads and fatigue
4. **Grid compliance:** Meet grid code requirements

5.6.2 Control Strategies

Below rated wind speed:

- Variable-speed operation
- Maximum power point tracking (MPPT)
- Optimal tip speed ratio maintenance

Above rated wind speed:

- Pitch control: Adjust blade angle to limit power
- Stall control: Allow aerodynamic stall
- Active stall: Combination of pitch and stall

5.6.3 Safety Systems

- Emergency shutdown systems
- Overspeed protection
- Vibration monitoring
- Lightning protection
- Fire detection and suppression
- Ice detection systems
- Parking brakes

Chapter 6

Wind Turbine Performance Analysis

6.1 Power Curve Analysis

The power curve represents turbine electrical output as a function of wind speed at hub height. It is the fundamental performance specification provided by manufacturers.

6.1.1 Power Curve Regions

1. **Cut-in to rated:** Power increases (approximately cubic relationship)
2. **Rated to cut-out:** Power maintained at rated value
3. **Cut-out:** Turbine shutdown for protection

6.1.2 Mathematical Representation

Empirical power curve models:

$$P(v) = \begin{cases} 0 & v < v_{cut-in} \\ P_r \cdot \frac{v^3 - v_{cut-in}^3}{v_r^3 - v_{cut-in}^3} & v_{cut-in} \leq v < v_r \\ P_r & v_r \leq v < v_{cut-out} \\ 0 & v \geq v_{cut-out} \end{cases}$$

where:

- P_r = Rated power
- v_r = Rated wind speed
- v_{cut-in} = Cut-in wind speed
- $v_{cut-out}$ = Cut-out wind speed

6.2 Tip Speed Ratio (TSR)

Tip speed ratio critically affects turbine efficiency:

$$\lambda = \frac{\omega R}{v} = \frac{2\pi n R}{60v}$$

where n is rotor speed in rpm.

6.2.1 Optimal TSR

Each turbine design has an optimal λ where C_P is maximized:

- Modern 3-blade turbines: $\lambda_{opt} = 7 - 8$
- 2-blade turbines: $\lambda_{opt} = 9 - 10$
- Multi-blade turbines: $\lambda_{opt} = 2 - 4$

Variable-speed control maintains λ_{opt} across varying wind speeds.

6.3 Cut-in, Rated, and Cut-out Wind Speeds

6.3.1 Cut-in Wind Speed

Minimum wind speed for electricity generation (typically 3-4 m/s). Below this speed, wind power insufficient to overcome system losses.

6.3.2 Rated Wind Speed

Wind speed at which turbine reaches rated (maximum continuous) power output (typically 12-15 m/s). Turbine operates at rated power for all wind speeds between rated and cut-out.

6.3.3 Cut-out Wind Speed

Maximum operational wind speed (typically 25 m/s). Above this, turbine shuts down to prevent damage. Modern turbines may have storm control allowing partial operation above traditional cut-out speeds.

6.4 Capacity Factor

Capacity factor (CF) measures actual energy production relative to theoretical maximum:

$$CF = \frac{E_{actual}}{P_r \times T}$$

where:

- E_{actual} = Actual energy produced
- P_r = Rated power
- T = Time period (typically 1 year)

Typical capacity factors:

- Onshore: 25-40%
- Offshore: 40-50%
- Exceptional sites: >50%

CF depends on:

- Wind resource quality
- Turbine design and rated power
- Availability and downtime
- Curtailment

6.5 Performance Monitoring and SCADA Systems

6.5.1 SCADA (Supervisory Control and Data Acquisition)

SCADA systems monitor and control turbine operations, collecting:

- Wind speed and direction
- Power output
- Rotor speed
- Generator temperature
- Vibration levels
- Pitch angles
- Grid parameters

Data resolution: typically 10-minute averages

6.5.2 Performance Metrics

Availability:

$$A = \frac{T_{\text{operational}}}{T_{\text{total}}} \times 100\%$$

Target availability: >95%

Performance ratio:

$$PR = \frac{E_{\text{actual}}}{E_{\text{predicted}}}$$

6.6 Losses and Efficiency Degradation

6.6.1 Energy Loss Categories

1. **Electrical losses** (1-3%):
 - Generator losses
 - Transformer losses
 - Collection system losses
2. **Environmental losses** (2-5%):
 - Blade soiling
 - Icing
 - High/low temperature shutdown
3. **Turbine performance** (2-5%):
 - Power curve degradation
 - Sub-optimal control
4. **Wake losses** (3-15%):
 - Wind farm internal wake effects
 - External wake effects
5. **Availability losses** (2-5%):
 - Scheduled maintenance
 - Unscheduled downtime
 - Grid unavailability
6. **Curtailement** (0-10%):
 - Grid constraints
 - Environmental restrictions
 - Economic curtailement

Total losses typically range from 10-25% depending on project specifics.

Chapter 7

Meteorological Data and Wind Measurement

7.1 Wind Measurement Equipment

7.1.1 Anemometers and Wind Vanes

Cup Anemometers:

- Most common for wind energy
- Rotating cups with angular velocity proportional to wind speed
- Accuracy: $\pm 1\text{-}2\%$ when calibrated
- Require regular maintenance

Sonic Anemometers:

- Measure wind using ultrasonic pulses
- No moving parts
- Higher accuracy and reliability
- More expensive
- Can measure 3D wind components

Wind Vanes:

- Measure wind direction
- Accuracy: $\pm 3\text{-}5^\circ$
- Essential for understanding directional distributions

7.1.2 Met Masts and Installation

Meteorological masts (met masts) are towers equipped with measurement instruments at multiple heights.

Design considerations:

- Height: Typically 60-100m, ideally matching hub height
- Measurement levels: Minimum 2-3 heights
- Boom orientation: Minimize tower shadow
- Redundancy: Multiple sensors at critical heights

Measurement campaign duration:

- Minimum: 1 year
- Recommended: 2-3 years
- Captures seasonal and inter-annual variability

7.1.3 Remote Sensing (LIDAR and SODAR)

LIDAR (Light Detection and Ranging):

- Uses laser pulses to measure wind profiles
- Ground-based or floating (offshore)
- Measures up to 200m+ height
- No physical structure at measurement height
- High accuracy (comparable to cup anemometers)
- Increasingly cost-competitive

SODAR (Sonic Detection and Ranging):

- Uses acoustic waves
- Measures up to 150m height
- Lower cost than LIDAR
- Sensitive to background noise
- Lower data recovery rates

7.2 Data Collection and Quality Control

7.2.1 Data Quality Assurance

Quality control procedures:

1. **Range checks:** Identify physically unrealistic values
2. **Stuck sensor detection:** Identify unchanging readings
3. **Icing detection:** Identify periods with ice-affected data
4. **Tower shadow effects:** Flag directional sectors with shadow
5. **Correlation checks:** Compare redundant sensors
6. **Shear validation:** Verify vertical profile consistency

7.2.2 Data Recovery Requirements

IEC 61400-12-1 standards specify:

- Minimum 12 months of data
- >90% data recovery rate
- Coverage of all wind directions
- Coverage of all wind speed bins

7.3 Meteorological Data Sources

7.3.1 Ground-Based Measurements

Advantages:

- Highest accuracy
- Site-specific data
- Direct measurement at hub height

Disadvantages:

- Expensive installation
- Time-consuming (minimum 1 year)
- Limited spatial coverage

7.3.2 Reanalysis Databases (NASA, ECMWF)

Reanalysis datasets combine historical observations with numerical weather models:

NASA MERRA-2:

- Spatial resolution: 50 km
- Temporal coverage: 1980-present
- Hourly time steps
- Global coverage

ECMWF ERA5:

- Spatial resolution: 30 km
- Temporal coverage: 1950-present
- Hourly time steps
- Generally considered highest quality reanalysis

Applications:

- Long-term wind resource assessment
- Inter-annual variability analysis
- Preliminary site screening
- Measure-Correlate-Predict reference data

7.3.3 Satellite Data

Satellite-based wind measurements:

- Limited to offshore/water surfaces
- Lower accuracy than ground measurements
- Useful for offshore resource screening
- Limited vertical resolution

7.4 Data Processing and Analysis

7.4.1 Data Filtering and Flagging

Process includes:

- Removal of invalid data points
- Flagging suspicious data
- Icing period identification
- Maintenance period exclusion

7.4.2 Statistical Analysis

Key analyses:

- Wind speed frequency distributions
- Directional distributions (wind roses)
- Diurnal (daily) patterns
- Seasonal patterns
- Wind shear profiles
- Turbulence intensity analysis

7.5 Measure-Correlate-Predict (MCP) Methods

MCP techniques extend short-term on-site measurements to long-term predictions using concurrent reference data.

7.5.1 MCP Methodology

1. Collect on-site measurements (1+ years)
2. Obtain long-term reference data (10+ years) for same period
3. Develop correlation model relating site to reference
4. Apply correlation to long-term reference period
5. Generate long-term wind resource estimate for site

7.5.2 MCP Techniques

Linear regression:

$$v_{site} = a + b \cdot v_{ref}$$

Matrix methods: Account for wind direction sectors

Advanced methods:

- Wind speed ratios
- Neural networks
- Weibull parameter scaling

7.6 Long-term Wind Resource Assessment

7.6.1 Inter-annual Variability

Wind resources vary year-to-year (typically $\pm 5\text{-}10\%$). Long-term assessment accounts for this variability to predict typical and extreme years.

7.6.2 P50/P90 Estimates

P50: Median (50% probability of exceedance) - typical year estimate

P90: Conservative (90% probability of exceedance) - used for financing

Difference between P50 and P90 typically 8-12%, reflecting inter-annual variability and uncertainty.

Chapter 8

Wind Farm Design and Optimization

8.1 Site Selection Criteria

Successful wind farm development begins with identifying sites that balance resource quality, accessibility, and constraints.

8.1.1 Primary Selection Factors

Wind Resource:

- Annual average wind speed >6 m/s at hub height
- Consistent wind directions
- Low turbulence intensity
- Favorable Weibull distribution parameters

Land Availability and Use:

- Sufficient contiguous area
- Compatible land uses (agriculture, grazing)
- Minimal residential proximity conflicts
- Land ownership and rights

Grid Connection:

- Proximity to transmission infrastructure
- Grid capacity and voltage level
- Interconnection costs
- Grid stability and strength

Accessibility:

- Existing road infrastructure

- Road upgrade requirements
- Transport routes for large components
- Distance from ports/manufacturing

Environmental and Regulatory:

- Environmental impact potential
- Protected species and habitats
- Permitting complexity
- Community acceptance
- Aviation and radar constraints

8.2 Wind Resource Mapping

8.2.1 Mapping Methodologies

Mesoscale modeling:

- Numerical weather prediction models
- Resolution: 1-10 km
- Captures large-scale features
- Examples: WRF, HARMONIE

Microscale modeling:

- Computational fluid dynamics (CFD)
- Resolution: 10-50 m
- Captures local terrain effects
- Examples: WAsP, WindPro, Openwind

Wind atlas methodology:

- Remove local effects from measurements
- Create generalized wind climate
- Apply to new locations with different terrain

8.2.2 Global Wind Atlas

The Global Wind Atlas provides worldwide preliminary resource assessment:

- Free online platform
- Mesoscale modeling (3 km resolution)
- Multiple height levels
- Useful for initial screening
- Should be verified with site measurements

8.3 Wind Farm Layout Optimization

Layout optimization seeks to maximize energy production while minimizing costs and wake losses.

8.3.1 Layout Design Objectives

- Maximize annual energy production (AEP)
- Minimize wake losses
- Minimize infrastructure costs
- Meet setback and environmental constraints
- Ensure adequate access for O&M

8.3.2 Optimization Approaches

Rule-based layouts:

- Regular grid patterns
- Typical spacing: $3\text{-}5D \times 5\text{-}9D$ (D = rotor diameter)
- Simple but often suboptimal

Gradient-based optimization:

- Local optimization methods
- Fast convergence
- Risk of local optima

Genetic algorithms:

- Global optimization approach
- Better handling of discrete variables
- Computationally intensive

8.4 Wake Effects and Wake Models

Upstream turbines create wakes (regions of reduced wind speed and increased turbulence), significantly impacting downstream turbines.

8.4.1 Wake Characteristics

Wake velocity deficit:

- Immediately downstream: 30-40% speed reduction
- Recovery distance: 5-15 rotor diameters
- Depends on atmospheric stability

Wake turbulence:

- Increased turbulence intensity in wake
- Affects downstream turbine loads
- Impacts fatigue lifetime

8.4.2 Wake Modeling Approaches

Engineering wake models:

Jensen (Park) model:

$$\frac{\Delta v}{v_0} = \frac{1 - \sqrt{1 - C_T}}{(1 + \alpha x/D)^2}$$

where:

- Δv = Velocity deficit
- C_T = Thrust coefficient
- α = Wake decay constant
- x = Downstream distance
- D = Rotor diameter

Eddy viscosity models: More sophisticated, account for wake evolution

CFD models: High-fidelity but computationally expensive

8.4.3 Wake Superposition

When turbine experiences multiple wakes, combination methods include:

- Linear summation
- Sum of squares
- Maximum deficit

8.5 Turbine Spacing and Micrositing

8.5.1 Spacing Guidelines

Typical spacing rules:

- **Perpendicular to prevailing wind:** 3-5 rotor diameters
- **Parallel to prevailing wind:** 5-9 rotor diameters
- Closer spacing increases wake losses but reduces infrastructure costs
- Wider spacing improves efficiency but requires more land

8.5.2 Micrositing Considerations

Fine-tuning individual turbine positions to:

- Maximize local wind resource
- Avoid excessive wake impacts
- Meet setback requirements
- Avoid environmentally sensitive areas
- Ensure construction and access feasibility

8.6 Array Efficiency

Array efficiency quantifies wake-related energy losses:

$$\eta_{array} = \frac{AEP_{actual}}{AEP_{free-stream}}$$

Typical values:

- Well-designed onshore farms: 90-95%
- Dense offshore arrays: 85-92%
- Poor layouts: <85%

8.7 Topography and Obstacle Considerations

8.7.1 Topographic Effects

Hilltops and ridges:

- Speed-up factors: 1.2-1.5×
- Excellent sites if accessible
- Increased turbulence over steep slopes

Valleys:

- Generally poor sites
- Sheltering effects reduce speeds
- Channeling can create favorable locations

8.7.2 Obstacles

Trees, buildings, and structures create:

- Wake zones: $5\text{--}20\times$ obstacle height
- Turbulence: affects region $2\text{--}5\times$ obstacle height upwind
- Setback requirements to minimize impacts

Chapter 9

Energy Production Estimation

9.1 Annual Energy Production (AEP) Calculation

AEP estimation is fundamental to project economics. The basic approach integrates wind speed distribution with turbine power curve.

9.1.1 Bin Method

Discretize wind speed into bins and sum energy contribution:

$$AEP = 8760 \times \sum_{i=1}^n f(v_i) \times P(v_i) \times \Delta v$$

where:

- $f(v_i)$ = Probability of wind speed bin i
- $P(v_i)$ = Power output at wind speed v_i
- Δv = Bin width
- 8760 = Hours per year

9.1.2 Weibull-Based AEP

Using Weibull distribution and power curve:

$$AEP = 8760 \int_0^{\infty} P(v) \cdot f(v) dv$$

where $f(v)$ is the Weibull probability density function.

For wind farms, calculate net AEP accounting for wake losses:

$$AEP_{farm} = \sum_{j=1}^N AEP_j \times (1 - L_{wake,j})$$

where:

- N = Number of turbines

- AEP_j = Gross AEP for turbine j
- $L_{wake,j}$ = Wake loss fraction for turbine j

9.2 Loss Factors and Availability

9.2.1 Comprehensive Loss Assessment

Table 9.1: Typical Energy Loss Factors

Loss Category	Typical Range (%)
Wake losses	3-15
Electrical losses	1-3
Availability losses	2-5
Turbine performance	2-5
Environmental (icing, soiling)	1-5
Curtailment	0-10
High wind hysteresis	0.5-2
Total typical losses	10-25

9.2.2 Net AEP Calculation

$$AEP_{net} = AEP_{gross} \times \prod_i (1 - L_i)$$

where L_i represents individual loss factors.

9.3 Uncertainty Analysis

AEP predictions contain multiple uncertainty sources that must be quantified for financial risk assessment.

9.3.1 Uncertainty Sources

Table 9.2: Typical AEP Uncertainty Components

Uncertainty Source	Typical Value (%)
Wind resource measurement	3-5
Long-term correction	3-6
Wind flow modeling	4-8
Power curve	3-5
Losses estimation	2-4
Inter-annual variability	5-10
Combined uncertainty (P50)	8-12

9.3.2 Combined Uncertainty

Assuming independent uncertainties:

$$\sigma_{total} = \sqrt{\sum_i \sigma_i^2}$$

9.4 P50/P90 Energy Estimates

9.4.1 Probability of Exceedance

P50 (Median):

- 50% probability actual production will exceed estimate
- Used for project valuation
- Best estimate of expected production

P90 (Conservative):

- 90% probability actual production will exceed estimate
- Used for debt financing
- Accounts for downside risk

9.4.2 P90 Calculation

Assuming normal distribution:

$$P90 = P50 \times (1 - 1.28 \times \sigma_{total})$$

where 1.28 is the z-score for 90th percentile.

Typical P90/P50 ratio: 0.85-0.92

9.5 Performance Ratio

Performance ratio compares actual to predicted energy:

$$PR = \frac{AEP_{measured}}{AEP_{predicted}} \times 100\%$$

9.5.1 PR Targets

- Excellent: >95%
- Good: 90-95%
- Acceptable: 85-90%
- Poor: <85%

Low PR indicates:

- Optimistic predictions
- Underperforming turbines
- Higher than expected losses
- Poor availability

9.6 Capacity Factor Analysis

9.6.1 Theoretical Maximum CF

For a given wind regime and turbine:

$$CF_{max} = \frac{1}{P_r} \int_0^{\infty} P(v) \cdot f(v) dv$$

9.6.2 Net Capacity Factor

$$CF_{net} = CF_{gross} \times (1 - L_{total})$$

9.6.3 Specific Yield

Energy per unit capacity:

$$SY = \frac{AEP_{net}}{P_r} \text{ (kWh/kW)}$$

Typical values:

- Onshore: 2,000-3,500 kWh/kW (CF: 23-40%)
- Offshore: 3,500-4,500 kWh/kW (CF: 40-51%)

Chapter 10

Wind Energy Economics and Project Development

10.1 Capital Costs (CAPEX)

Capital expenditure encompasses all costs to develop and construct a wind farm.

10.1.1 CAPEX Breakdown

Table 10.1: Typical CAPEX Distribution (Onshore)

Component	Percentage of Total
Turbines (supply and installation)	60-70%
Balance of system	15-25%
- Civil works and foundations	5-10%
- Electrical infrastructure	5-10%
- Grid connection	3-8%
Development and consenting	3-8%
Engineering and management	3-5%
Financial costs	2-5%
Contingency	3-5%

10.1.2 Specific CAPEX

Cost per unit capacity:

$$CAPEX_{specific} = \frac{Total\ CAPEX}{Installed\ Capacity\ (MW)}$$

Typical ranges:

- Onshore: \$1,200-1,800/kW
- Offshore: \$3,000-5,000/kW

Note: Costs vary significantly by region, project size, and site characteristics.

10.2 Operational and Maintenance Costs (OPEX)

10.2.1 OPEX Components

Fixed OPEX:

- Scheduled maintenance contracts
- Insurance premiums
- Land lease payments
- Property taxes
- Administration and management

Variable OPEX:

- Unscheduled repairs
- Component replacements
- Performance-based payments

10.2.2 OPEX Estimates

Typical annual OPEX:

- Onshore: \$40-70/kW/year
- Offshore: \$80-150/kW/year

OPEX often increases with turbine age due to:

- Warranty expiration
- Component wear
- Increased failure rates

10.3 Levelized Cost of Energy (LCOE)

LCOE represents average cost per unit energy over project lifetime, enabling comparison across technologies and projects.

10.3.1 LCOE Formula

$$LCOE = \frac{\sum_{t=0}^n \frac{I_t + M_t}{(1+r)^t}}{\sum_{t=1}^n \frac{E_t}{(1+r)^t}}$$

where:

- I_t = Investment expenditure in year t

- M_t = Operations and maintenance expenditure in year t
- E_t = Energy production in year t
- r = Discount rate
- n = Project lifetime

10.3.2 Simplified LCOE

$$LCOE = \frac{CAPEX \times CRF + OPEX_{annual}}{AEP_{net}}$$

where CRF (Capital Recovery Factor) is:

$$CRF = \frac{r(1+r)^n}{(1+r)^n - 1}$$

10.3.3 Current LCOE Values

Global benchmarks (2024):

- Onshore wind: \$30-60/MWh
- Offshore wind: \$70-120/MWh

Morocco context: Competitive with fossil fuels, particularly in high-resource areas.

10.4 Financial Analysis and NPV

10.4.1 Net Present Value

$$NPV = \sum_{t=0}^n \frac{CF_t}{(1+r)^t}$$

where:

- CF_t = Net cash flow in year t
- r = Discount rate
- n = Project lifetime

Decision criteria:

- $NPV > 0$: Project is economically viable
- $NPV < 0$: Project destroys value

10.4.2 Cash Flow Components

Revenue:

$$Revenue_t = AEP_t \times Price_{electricity,t}$$

Costs:

- CAPEX (year 0)
- Annual OPEX
- Major overhaul costs
- Decommissioning costs

Taxes and depreciation:

- Corporate income tax
- Tax depreciation benefits

10.5 Payback Period and IRR

10.5.1 Payback Period

Time required to recover initial investment:

Simple payback ignores time value of money:

$$Payback = \frac{CAPEX}{Annual\ Net\ Cash\ Flow}$$

Typical payback periods:

- Without subsidies: 8-15 years
- With subsidies: 5-10 years

10.5.2 Internal Rate of Return (IRR)

Discount rate where NPV = 0:

$$0 = \sum_{t=0}^n \frac{CF_t}{(1 + IRR)^t}$$

Target IRR depends on:

- Project risk profile
- Financing structure
- Investor requirements

Typical targets:

- Low-risk projects: 6-10%
- Moderate-risk: 10-15%
- High-risk: >15%

10.6 Incentives and Policy Frameworks

10.6.1 Support Mechanisms

Feed-in Tariffs (FIT):

- Guaranteed price per kWh
- Long-term contracts (15-20 years)
- Provides revenue certainty

Renewable Energy Certificates (RECs):

- Tradeable certificates for renewable generation
- Additional revenue stream
- Market-based mechanism

Tax incentives:

- Production tax credits
- Investment tax credits
- Accelerated depreciation
- Import duty exemptions

Morocco-specific incentives:

- MASEN programs
- Competitive tender processes
- Grid connection support
- Land concessions

10.7 Grid Integration Costs

10.7.1 Connection Costs

- Substation construction/upgrade
- Transmission line extension
- Protection and control systems
- Metering and communication

Costs highly site-specific, ranging from \$50-300/kW depending on distance and required upgrades.

10.7.2 Grid Services

Modern wind farms provide:

- Reactive power support
- Voltage regulation
- Frequency response
- Black start capability (some systems)

These capabilities may command additional revenue or reduce integration costs.

Chapter 11

Introduction to System Advisor Model (SAM)

11.1 Overview of SAM Software

11.1.1 What is SAM?

The System Advisor Model (SAM) is a comprehensive performance and financial modeling software developed by the National Renewable Energy Laboratory (NREL) in the United States. SAM enables users to evaluate renewable energy projects, particularly solar photovoltaic and wind power systems, by simulating technical performance and conducting detailed financial analysis.

Key Characteristics:

- Free, open-source software
- Developed and maintained by NREL
- Regular updates and improvements
- Extensive documentation and support
- Cross-platform compatibility (Windows, Mac, Linux)

11.1.2 SAM Capabilities for Wind Energy

SAM's wind power module provides comprehensive modeling capabilities:

Performance Modeling:

- Single turbine or wind farm simulation
- Hour-by-hour energy production calculation
- Multiple wake loss models
- Weibull distribution or time-series wind data
- Power curve modeling and customization

Financial Modeling:

- Multiple financial structures (residential, commercial, utility)
- Cash flow analysis
- LCOE calculation
- NPV and IRR computation
- Tax and incentive modeling

Advanced Features:

- Parametric analysis
- Sensitivity analysis
- Stochastic simulation
- P50/P90 analysis
- Python scripting (PySAM)

11.1.3 SAM vs Other Wind Energy Software

Table 11.1: Wind Energy Software Comparison

Feature	SAM	WindPRO	OpenWind
Cost	Free	Commercial	Commercial
Financial modeling	Excellent	Good	Limited
Wind flow modeling	Basic	Advanced	Advanced
Wake modeling	Good	Excellent	Excellent
Layout optimization	Basic	Advanced	Advanced
Learning curve	Moderate	Steep	Steep
Documentation	Excellent	Good	Good

When to use SAM:

- Techno-economic feasibility studies
- Financial optimization
- Academic research and education
- Preliminary project assessment
- Budget-conscious projects

11.2 Getting Started with SAM

11.2.1 Installation and Setup

Download and Installation:

1. Visit: <https://sam.nrel.gov>
2. Download latest version for your operating system
3. Run installer and follow prompts
4. Accept license agreement
5. Complete installation

System Requirements:

- Operating System: Windows 10/11, macOS 10.13+, Linux
- RAM: 4 GB minimum, 8 GB recommended
- Disk Space: 500 MB
- Display: 1280×1024 minimum resolution

11.2.2 User Interface Navigation

Main Interface Components:

1. **Menu Bar:** File operations, preferences, help
2. **Navigation Menu (left):** Access input pages
3. **Input Panel (center):** Parameter entry
4. **Results Section:** Simulation outputs
5. **Messages:** Warnings and error notifications

Input Page Organization:

- Location and Resource
- Wind Turbine
- Wind Farm
- System Costs
- Financial Parameters

11.2.3 Creating a New Project

Step-by-step Process:

1. Launch SAM
2. Click "Start a new project"
3. Select technology: "Wind Power"
4. Choose financial model:
 - Single Owner (utility-scale)
 - Commercial
 - Residential
 - None (performance only)
5. Name your project
6. Begin entering inputs

11.3 Wind Resource Input in SAM

11.3.1 Wind Resource Data File Format

SAM accepts wind data in specific formats:

File Types:

- SAM CSV format (comma-separated)
- SRW format (Solar Resource for Wind)
- TMY formats

Required Data Fields:

- Timestamp (year, month, day, hour)
- Wind speed at measurement height (m/s)
- Wind direction (degrees)
- Temperature (°C)
- Pressure (atm or mbar)

Example CSV Structure:

```
Year,Month,Day,Hour,Speed,Direction,Temperature,Pressure
2020,1,1,1,7.5,245,15.2,1.013
2020,1,1,2,7.8,250,15.0,1.012
...
```

11.3.2 Downloading Wind Data (NREL, NASA)

NREL Wind Toolkit:

1. Visit: <https://www.nrel.gov/grid/wind-toolkit.html>
2. Select location on map
3. Choose temporal resolution
4. Download data in SAM-compatible format

NASA POWER:

1. Visit: <https://power.larc.nasa.gov>
2. Enter coordinates
3. Select parameters
4. Download data
5. Convert to SAM format if necessary

11.3.3 Weibull Distribution Input

Alternative to time-series data, SAM accepts Weibull parameters:

Required Inputs:

- Weibull k (shape parameter)
- Average wind speed or Weibull c (scale parameter)
- Reference height
- Wind shear exponent

When to use Weibull:

- Limited measured data available
- Preliminary assessment
- Simplified analysis
- Comparison with time-series results

11.3.4 Wind Shear Coefficient

SAM uses power law to extrapolate wind speeds to hub height:

Input Options:

- Default value (0.14 for open terrain)
- Site-specific value from measurements
- Typical values by terrain type

Impact on Results:

- Higher shear coefficient $\rightarrow$ greater speed increase with height
- Significant impact on AEP for tall turbines
- Should be validated with multi-height measurements

11.4 Wind Turbine Specification in SAM

11.4.1 Selecting Turbines from Library

SAM includes an extensive turbine library:

Accessing Library:

1. Navigate to "Wind Turbine" page
2. Select "Choose from library"
3. Browse or search manufacturers
4. Select turbine model
5. Review specifications

Library Information Includes:

- Manufacturer and model
- Rated power (kW or MW)
- Rotor diameter (m)
- Hub height options (m)
- Power curve data
- Cut-in, rated, cut-out speeds

11.4.2 Defining Custom Turbine Characteristics

For turbines not in library or customized analysis:

Option 1: Design Using Turbine Characteristics

- Rated power
- Rotor diameter
- Hub height
- Drive train efficiency
- Maximum tip speed

SAM calculates power curve using NREL's model.

Option 2: Enter Power Curve Directly

- Input wind speed vs. power table
- Specify cut-in and cut-out speeds
- Define hub height
- Enter rotor diameter

11.4.3 Power Curve Input and Calculation

Power Curve Table Format:

Table 11.2: Example Power Curve Entry

Wind Speed (m/s)	Power (kW)
0-3	0
4	50
5	150
...	...
12	2000
13-25	2000
>25	0

Power Curve Adjustments:

- Air density correction
- Temperature effects
- Altitude compensation

11.4.4 Hub Height and Rotor Diameter

Hub Height Selection:

- Determines wind speed via shear extrapolation
- Available heights from manufacturer
- Custom height input possible
- Higher hub heights generally increase production

Rotor Diameter:

- Determines swept area: $A = \pi R^2$
- Affects power capture
- Impacts wake calculations
- Used in turbine spacing design

11.5 Wind Farm Layout Configuration

11.5.1 Single Turbine vs Wind Farm

Single Turbine Mode:

- Simplest configuration
- No wake losses
- Suitable for distributed wind
- Useful for turbine comparison studies

Wind Farm Mode:

- Multiple turbines
- Wake loss calculations
- Layout specification required
- Realistic project modeling

11.5.2 Farm Size Specification

Approximate Farm Size Option:

- Specify desired total capacity (MW)
- SAM calculates number of turbines
- Assumes generic layout
- Quick preliminary analysis

Example:

- Turbine: 2.5 MW
- Desired farm size: 50 MW
- Number of turbines: $50/2.5 = 20$ turbines

11.5.3 Detailed Layout Design

Coordinate-Based Layout:

- Specify X-Y coordinates for each turbine
- More realistic wake modeling
- Accounts for actual positioning
- Requires layout design beforehand

Input Format:

Table 11.3: Turbine Coordinate Table

Turbine ID	X (m)	Y (m)
1	0	0
2	500	0
3	0	700
...	...	...

11.5.4 Wake Loss Models (Simple, Park, Eddy Viscosity)

Simple Wake Model:

- Applies uniform wake loss percentage
- No directional considerations
- Fastest computation
- Least accurate

Park (Jensen) Wake Model:

- Industry-standard engineering model
- Accounts for wake expansion
- Directional wake calculations
- Moderate computational requirements
- Recommended for most applications

Eddy Viscosity Wake Model:

- More sophisticated physics
- Better wake recovery prediction
- Higher computational cost
- More accurate for complex layouts

Model Selection Guidance:

- Preliminary studies: Simple
- Standard analysis: Park
- Detailed studies: Eddy Viscosity

11.6 System Losses and Availability

11.6.1 Electrical Losses

Loss Components:

- Array cable losses (turbine to substation)
- Transformer losses
- Transmission losses (substation to grid)

Typical Values:

- Collection system: 1-2%
- Transmission: 0.5-1.5%
- Total electrical: 2-3%

11.6.2 Environmental Losses

Loss Types:

- **Icing:** Blade ice accumulation (0-5% depending on climate)
- **Blade soiling:** Dirt, dust accumulation (0.5-2%)
- **Extreme temperatures:** Shutdown during extreme conditions (0-2%)
- **High winds:** Shutdown above cut-out speed (accounted in power curve)

11.6.3 Availability and Curtailment

Availability Loss:

$$L_{availability} = 1 - \frac{Operational\ Hours}{Total\ Hours}$$

Components:

- Scheduled maintenance (1-2%)
- Unscheduled outages (1-3%)
- Grid unavailability (0-1%)

Target availability: 95-98%

Curtailment:

- Grid constraints
- Negative pricing periods
- Environmental restrictions (bird migration, noise)
- Typical: 0-5%

11.7 Cost Modeling in SAM

11.7.1 NREL Cost and Scaling Model

SAM includes built-in cost models based on NREL research:

Turbine Cost Calculation:

$$Cost_{turbine} = f(Rated\ Power, Rotor\ Diameter, Hub\ Height)$$

SAM estimates using empirical correlations from industry data.

Advantages:

- Quick estimates without detailed quotes
- Regularly updated based on market trends
- Useful for preliminary feasibility

Limitations:

- Generic costs may not reflect specific quotes
- Regional variations not fully captured
- Should be validated with actual project costs

11.7.2 Balance of System Costs

BOS Components:

- Site preparation and civil works
- Foundations
- Electrical infrastructure
- Substation
- Grid interconnection
- Access roads
- Project management

Input Options:

- Use SAM's detailed BOS model (many inputs)
- Enter total BOS cost directly
- Specify cost per kW

11.7.3 O&M Cost Inputs

O&M Input Methods:

1. Fixed Annual Cost:

$$OPEX = \$/kW/year \times Capacity$$

2. Variable Cost:

$$OPEX = \$/MWh \times Annual Production$$

3. Detailed Component Costs:

- Labor costs
- Spare parts
- Insurance
- Land lease
- Administration

Escalation: Allow for O&M cost increases over project life (typically 1-2%/year above inflation).

11.8 Financial Modeling

11.8.1 Selecting Financial Models

SAM offers multiple financial structures:

Single Owner:

- Utility-scale projects
- Corporate ownership
- Merchant or PPA revenue
- Debt and equity financing

Commercial:

- Behind-the-meter projects
- Retail electricity offset
- Demand charge reduction
- Third-party ownership options

Residential:

- Small-scale systems
- Homeowner perspective
- Simple financing
- Net metering

None (Performance Only):

- No financial calculations
- Energy production only
- Technical assessment

11.8.2 Electricity Rate Structures

For Utility-Scale (Single Owner):

Power Purchase Agreement (PPA):

- Fixed price (\$/MWh)
- Escalation rate
- Contract term
- Price floor/ceiling

Merchant/Market Sales:

- Market price forecast
- Price volatility
- Revenue risk

**For Behind-the-Meter (Commercial):
Retail Rate Structure:**

- Energy charges (\$/kWh)
- Demand charges (\$/kW)
- Time-of-use periods
- Seasonal variations

11.8.3 Incentives and Tax Credits

Investment Tax Credit (ITC):

- Percentage of CAPEX
- Affects depreciation basis
- US-specific but similar mechanisms exist globally

Production Tax Credit (PTC):

- Payment per kWh produced
- Typically first 10 years
- Inflation-adjusted

Depreciation:

- MACRS (US) or local equivalents
- Accelerated depreciation schedules
- Tax benefit timing

Other Incentives:

- Grants and rebates
- Preferential loans
- Feed-in tariffs
- Renewable energy certificates

11.9 Running Simulations and Results

11.9.1 Annual Energy Production

After inputting all parameters, SAM calculates:

Gross Energy Production:

- Before losses
- Individual turbine production
- Total farm production

Net Energy Production:

- After all losses applied
- Delivered to grid
- Basis for revenue calculations

Output Resolution:

- Annual totals
- Monthly averages
- Hourly time series (8,760 hours)

11.9.2 Performance Metrics

Key Performance Indicators:

- Annual Energy Production (MWh)
- Capacity Factor (%)
- Specific Yield (kWh/kW)
- Performance Ratio (%)

Loss Summary:

- Wake losses
- Electrical losses
- Environmental losses
- Availability losses
- Total losses

11.9.3 Financial Metrics (NPV, LCOE, Payback)

Levelized Cost of Energy:

- Real LCOE (constant dollars)
- Nominal LCOE (actual dollars)
- Comparison benchmark

Net Present Value:

- Total project NPV
- Sensitivity to discount rate
- Investment decision criterion

Internal Rate of Return:

- Project IRR
- Comparison to hurdle rate
- Return on investment indicator

Payback Periods:

- Simple payback
- Discounted payback
- Time to positive cash flow

11.9.4 Monthly and Hourly Results

Monthly Results:

- Seasonal production patterns
- Monthly revenue
- Monthly capacity factors
- Useful for operational planning

Hourly Results:

- Complete 8,760-hour time series
- Export to CSV for detailed analysis
- Integration with other models
- Correlation with demand profiles

11.10 Advanced Features in SAM

11.10.1 Parametric Analysis

Systematically vary one or more inputs to assess impact on results:

Use Cases:

- Hub height optimization
- Turbine size comparison
- Financial structure sensitivity
- Electricity price impact

Setup:

1. Select "Parametrics" from main menu
2. Choose parameter(s) to vary
3. Define range and steps
4. Select output metrics
5. Run parametric sweep
6. Analyze results in tables/graphs

11.10.2 Sensitivity Analysis

Quantify impact of input uncertainties on key outputs:

Process:

1. Define uncertain parameters
2. Specify uncertainty ranges
3. Run sensitivity analysis
4. Identify most influential parameters

Applications:

- Risk assessment
- Parameter prioritization
- Data collection guidance
- Uncertainty quantification

11.10.3 P50/P90 Analysis

Statistical analysis of expected production:

Methodology:

- Define probability distributions for uncertain inputs
- Monte Carlo simulation
- Generate distribution of AEP
- Calculate P50, P90, P99 values

Critical for:

- Project financing
- Risk quantification
- Conservative estimates
- Investor confidence

11.10.4 Scripting with Python (PySAM)

PySAM provides Python interface to SAM's simulation engine:

Capabilities:

- Automate repetitive analyses
- Integrate SAM with other tools
- Custom optimization algorithms
- Batch processing
- Advanced data analysis

Example Use Cases:

- Multi-site assessment
- Layout optimization
- Custom financial models
- Integration with GIS

Getting Started with PySAM:

```
pip install NREL-PySAM
import PySAM.Windpower as wp
model = wp.default("WindPowerSingleOwner")
# Set inputs and run simulation
```

11.11 Practical SAM Tutorial

11.11.1 Case Study: Modeling a Wind Farm in Morocco

Project Specifications:

- Location: Tangier region, Morocco
- Site coordinates: 35.7°N, 5.9°W
- Capacity: 50 MW
- Turbine: 2.5 MW, 100m rotor diameter, 80m hub height
- Number of turbines: 20
- Average wind speed: 7.5 m/s at 80m

11.11.2 Step-by-Step Simulation Workflow

Step 1: Create New Project

1. Open SAM
2. File → New
3. Select "Wind Power"
4. Select "Single Owner" financial model
5. Name: "Morocco_Tangier_50MW"

Step 2: Wind Resource Input

1. Navigate to "Location and Resource"
2. Option A: Upload wind data file
 - Download data from NASA POWER or NREL
 - Format as SAM CSV
 - Upload file
3. Option B: Use Weibull distribution
 - Average wind speed: 7.5 m/s
 - Weibull k: 2.0 (typical)
 - Reference height: 80m
 - Shear coefficient: 0.14

Step 3: Turbine Selection

1. Navigate to "Wind Turbine"
2. Select from library or define custom turbine:

- Rated power: 2,500 kW
- Rotor diameter: 100 m
- Hub height: 80 m
- Cut-in speed: 3 m/s
- Rated speed: 12 m/s
- Cut-out speed: 25 m/s

Step 4: Wind Farm Configuration

1. Navigate to "Wind Farm"
2. Select "Specify farm size"
3. Enter: 50 MW
4. SAM calculates: 20 turbines
5. Select wake model: "Park (Jensen)"
6. Set wake decay coefficient: 0.075 (default)
7. Enter losses:
 - Electrical: 2%
 - Availability: 3%
 - Environmental: 2%
 - Other: 1%

Step 5: Cost Inputs

1. Navigate to "System Costs"
2. Turbine costs:
 - Use NREL model or enter: \$1,400/kW
 - Total turbine cost: \$70 million
3. Balance of system: $\$400/\text{kW} = \20 million
4. Total CAPEX: \$90 million
5. Annual O&M: $\$50/\text{kW}/\text{year} = \$2.5 \text{ million}/\text{year}$

Step 6: Financial Parameters

1. Navigate to "Financial Parameters"
2. Analysis period: 25 years
3. PPA price: \$60/MWh
4. PPA escalation: 2%/year

5. Discount rate: 8%
6. Debt fraction: 70%
7. Debt interest rate: 6%
8. Debt term: 18 years

Step 7: Run Simulation

1. Click "Simulate" button
2. Review messages for errors/warnings
3. Wait for completion

11.11.3 Results Interpretation and Optimization

Example Results:

Performance:

- Gross AEP: 165,000 MWh
- Net AEP: 152,000 MWh
- Capacity factor: 35%
- Total losses: 8%

Financial:

- LCOE: \$52/MWh
- NPV: \$15 million
- IRR: 11.5%
- Payback: 10 years

Optimization Opportunities:

1. **Hub Height:** Run parametric varying 80-120m
2. **Turbine Size:** Compare 2.0 vs 2.5 vs 3.0 MW
3. **Layout:** Optimize spacing to reduce wake losses
4. **Financial Structure:** Adjust debt/equity ratio

Sensitivity Analysis:

- Wind speed ($\pm 10\%$): $\pm 20\text{-}30\%$ impact on NPV
- PPA price ($\pm \$10/\text{MWh}$): $\pm 15\text{-}20\%$ impact on NPV
- CAPEX ($\pm 10\%$): $\pm 5\text{-}8\%$ impact on IRR
- O&M ($\pm 20\%$): $\pm 3\text{-}5\%$ impact on LCOE

Conclusion

This comprehensive course has covered the fundamental principles of wind energy systems, from atmospheric physics and turbine aerodynamics to economic analysis and SAM software proficiency. The System Advisor Model provides a powerful, accessible platform for conducting detailed wind energy assessments, enabling students and professionals to make informed decisions about wind power projects.

Key Takeaways

- Wind energy conversion involves complex interplay of meteorology, aerodynamics, mechanics, and economics
- Accurate resource assessment and loss characterization are critical for reliable predictions
- SAM offers comprehensive techno-economic modeling capabilities suitable for various project types
- Practical application and iteration are essential for developing proficiency

Next Steps for Students

1. Practice with SAM using different project scenarios
2. Explore SAM's advanced features and scripting capabilities
3. Apply learned concepts to real wind farm case studies
4. Continue learning through POWERUP Club's additional simulation software training
5. Engage with the wind energy research community

Thank you for participating in this POWERUP Club formation!

*For questions, suggestions, or collaboration:
Contact POWERUP Club - ENSA Fes*